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B. E. (Sixth Semester) Examination,Nov.-Dec. 2016

(New Scheme)

(Elect. Branch)

ELECTRICAL MACHINES - III

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Part (a) of each question is compulsory. Attempt any two parts from (b), (c) & (d).*

1 (a) What is effect of additional rotor circuit in synchronous machine? 2

(b) Show that the relation between power and torque :

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$$T = \frac{3}{2} \times \frac{P}{2} (\lambda_d i_q - \lambda_q i_d)$$

(c) Derive the steady state power angle equation and characteristics for cylindrical rotor and synchronous machine. 7

(d) With usual notation prove that in 3ϕ, 2 pole synchronous machine 7

$$L_{aa} = L_{ao} + L_{ag} + L_{ag} \cos 2\theta$$

2 (a) Write down only transformation matrix into dqo variables.2

(b) A 400 HP, 440V (L – L), 60 CPS, γ connected 6 pole SCIM has a full load efficiency of 93% and power factor of 90%. The motor constant in ohms per phase referred to stator or as follows

$x_1 = 0.060$, $x_2 = 0.060$, $x_\phi = 2.50$

$r_1 = 0.0073$, $r_2 = 0.0064$

While the motor is operating in the steady state under rated condition, a 3 phase short circuit occur on its supply line near to motor terminal. Determine motor rms short circuit current. 7

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(c) Derive an expression for transformation to be power invarient using part transformation. 7

(d) From d and q-axis voltage expression, derive the equivalent circuit of 3 phase induction motor for balanced input. 7

3 (a) Explain why single phase induction motors are not self starting. 2

(b) Two phase induction motor with stator voltage V_r and V_c which are unequal but in space quadrature, produces some starting torque. Show that same startingh torque is produced by two phase balanced voltage magnitude is $\sqrt{V_r V_c}$. 7

(c) A 230V, 4-pole, 50 CPS, single phase induction motor has the following constant and losses. 7

r_1 = 2.3 ohm, x_1 = 3.2 ohm,

r_2 = 4.2 ohm, x_2 = 3.2 ohm

x_m = 74 ohm, core loss = 98 watts

friction and windage loss = 30 watts.

If this motor is running at slip of 0.05 at rated voltage and

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frequency, find the stator current, power factor, output power.

(d) Derive the equivalent circuit of a single phase induction motor with the help of double revoluing field theory. 7

4 (a) Which two types emf induced in commutator winding. 2

(b) Describe the improving commutation in a single phase ac series motor 7

(c) Draw the phasor diagram and operating characteristics of AC series motor. 7

(d) A 2 pole ac commulator machine has the following data 7

Armature turn = 300

Armature speed = 1800 rpm

Frequency and maximum value of pulsating flux = 50 Hz and 0.02 wb respectively find the magnitude and frequency of emf appearing at the brushes, If the brush axis is

(i) 90° away from field axis.

(ii) aligned along the field axis.

(iii) 30° away from field axis 7

5 (a) What is the application of stepper motor give any-four. 2

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(b) Explain the construction and working principle of reluctance motor. 7

(c) Describe the three pulse brushless DC motor. How-torque is developed in the motor. 7

(d) Explain the construction and working principle of linear induction motor. 7

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